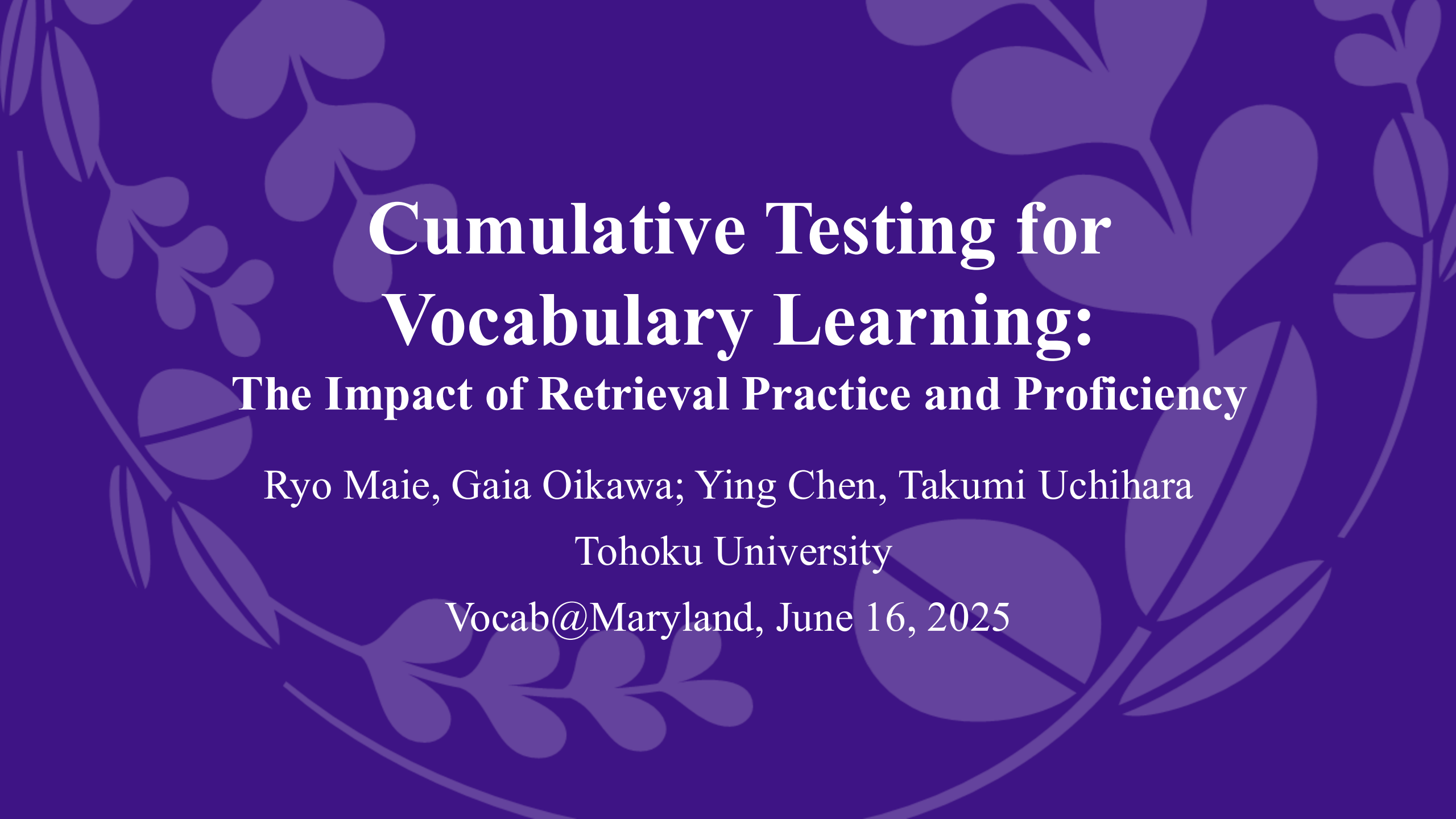




Cumulative Testing for Vocabulary Learning:

The Impact of Retrieval Practice and Proficiency

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Vocab@Maryland, June 16, 2025

Testing and Retrieval Practice Effect

Vocabulary practice and testing

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Testing effect (aka: retrieval practice effect) (Roediger & Karpicke, 2006)

- Testing (retrieval) leads to more potent learning than mere studying (e.g., paired-associate learning)

What's does this word mean in English?

pájaro = _____

vs.

Memorize the association of the Spanish-English- pair

pájaro = bird

- **Retrieval** (compared to studying)
 - greater effort and hence stronger memory trace (Roediger & Butler, 2011)
 - desirable difficulty
- **L2 vocabulary learning**
 - recall leads to greater learning of new vocabulary (Barcroft, 2007)

Barcroft, J. (2007). Effects of opportunities for word retrieval during second language vocabulary learning. *Language Learning*, 57(1), 35–56.

Roediger, H. L. III, & Butler, A. C. (2011). The critical role of retrieval practice in long-term retention. *Trends in Cognitive Sciences*, 15(1), 20–27.

Roediger, H. L. III, & Karpicke, J. D. (2006). Test-enhanced learning: Taking memory tests improves long-term retention. *Psychological Science*, 17(3), 249–255.

Cumulative testing

- testing not only recent material but also previously studied material



100 target words

- 10 words introduced each week
 1. 10 words (W1: 10 words)
 2. 10 words (W1: 5 words & W2: 5 words)
 3. 10 words (W1: 2 & W2: 3 & W3: 5 words)
 4. 10 words (W1: 1 & W2: 2 & W3: 2 & W4: 5)

Superior learning

- psychology (Lawrence, 2013)
- mathematics (Beagley & Capaldi, 2016)
- biology (Bailey et al., 2021)
- language (Kanayama et al., 2023 ; Nakata et al., 2021)



Bailey, E. G., Greenall, R. F., Tullis, M. M., & Williams, K. R. (2021). The retention benefits of cumulative versus non-cumulative midterms in introductory biology may depend on students' reasoning skills. *PLOS ONE*, 16(4), e0250143.

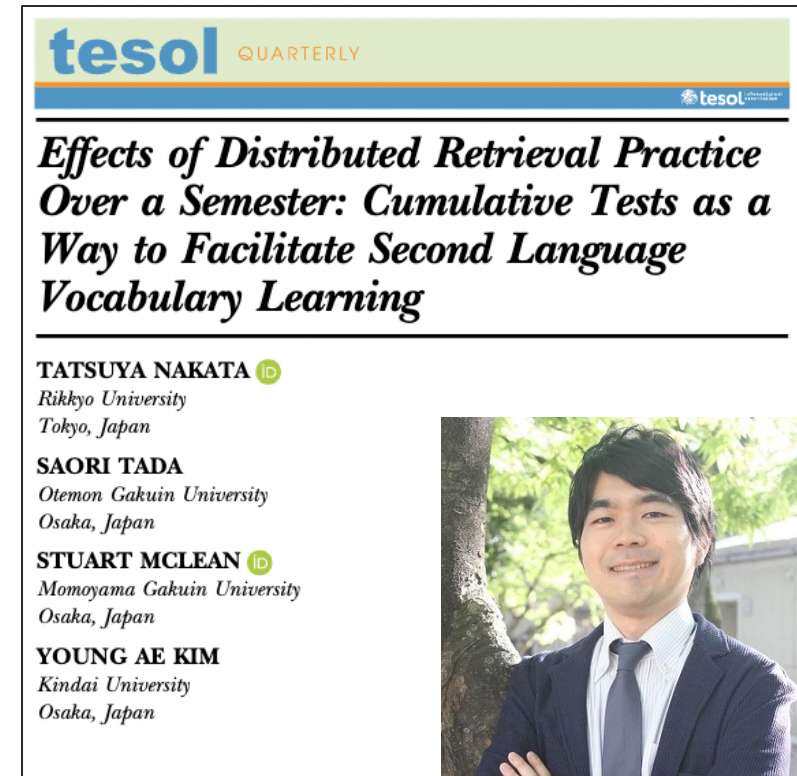
Beagley, J. E., & Capaldi, M. (2016). The effect of cumulative tests on the final exam. *PRIMUS: Problems, Resources, and Issues in Mathematics Undergraduate Studies*, 26(9), 878–888.

Kanayama, K., Iwata, A., & Kasahara, K. (2023). Effects of random selection tests on second language vocabulary learning: A comparison with cumulative tests. *International Review of Applied Linguistics in Language Teaching*.

Lawrence, N. K. (2013). Cumulative exams in the introductory psychology course. *Teaching of Psychology*, 40(1), 15–19.

Cumulative vs. Noncumulative for vocabulary

- University students in Japan
 - majoring in pharmacy
- 80 medical words
- 8 weekly quizzes
- pretest / posttest
 - meaning recall (L2 to L1 translation)
 - form recall (L1 to L2 translation)



Class	Target items introduced	Weekly quiz	Dependent measures
1	Item set 1		Pretest
2	Item set 2	Weekly quiz 1	
3	Item set 3	Weekly quiz 2	
4	Item set 4	Weekly quiz 3	
5	Item set 5	Weekly quiz 4	
6	Item set 6	Weekly quiz 5	
7	Item set 7	Weekly quiz 6	
8	Item set 8	Weekly quiz 7	
9		Weekly quiz 8	
10			
11			
12			Posttest

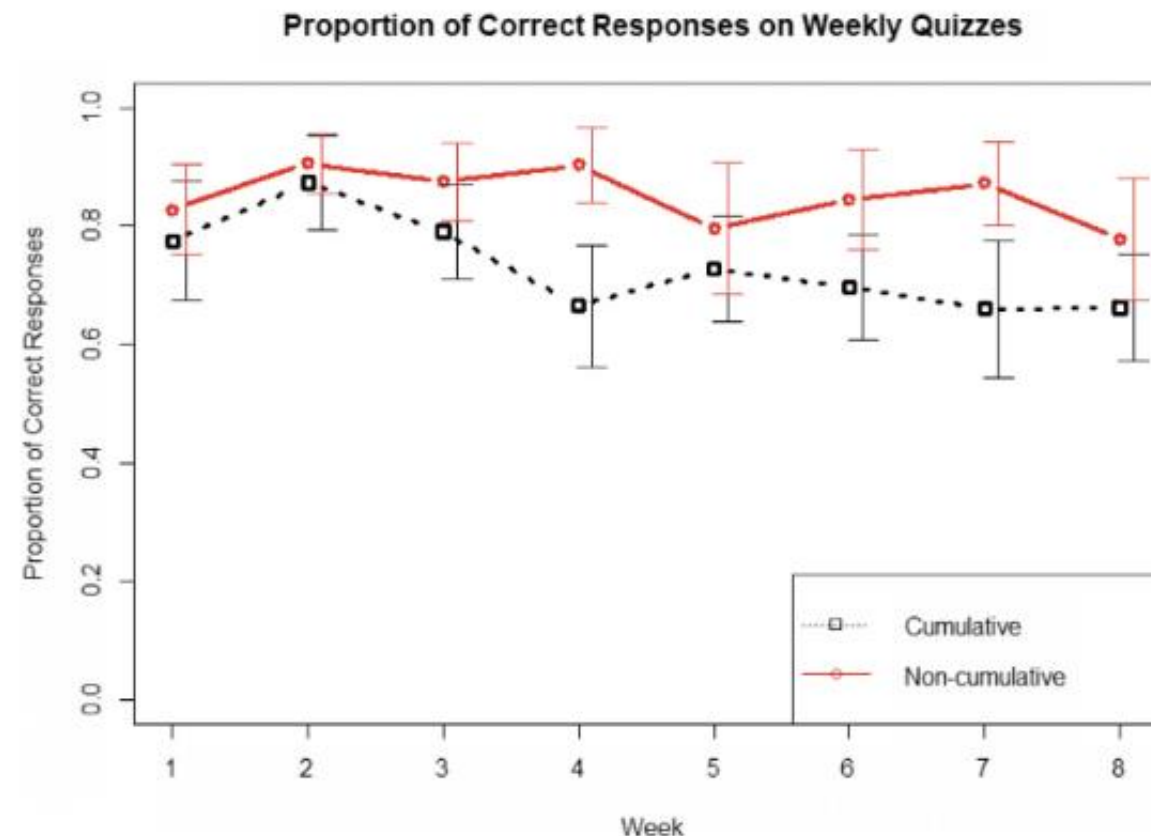


FIGURE 2. Proportion of correct responses (%) on the weekly quizzes. Error bars indicate 95% confidence intervals. [Colour figure can be viewed at [wileyonlinelibrary.com](https://onlinelibrary.wiley.com)]

TABLE 1
Proportion of Correct Responses on the Pretest and Posttest

		Noncumulative		Cumulative	
		<i>M</i>	<i>SD</i>	<i>M</i>	<i>SD</i>
Productive	Pretest	1.6% [0.6%, 2.7%]	3.0%	3.1% [1.7%, 4.4%]	4.0%
	Posttest	12.1% [8.8%, 15.5%]	9.8%	32.5% [29.9%, 47.2%]	25.7%
	Gain	10.5% [7.2%, 13.8%]	9.6%	35.5% [27.2%, 43.8%]	24.7%
Receptive	Pretest	4.3% [2.8%, 5.8%]	4.4%	5.3% [3.6%, 6.9%]	4.8%
	Posttest	26.4% [21.7%, 31.1%]	13.8%	50.9% [43.1%, 58.6%]	23.1%
	Gain	22.1% [18.1%, 26.1%]	11.8%	45.6% [38.2%, 53.0%]	22.1%

Note. Values inside brackets indicate 95% confidence intervals. Gain = posttest score – pretest score.

Some questions inspired by Nakata et al. (2021, *TQ*)

Cumulative vs. Noncumulative testing

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1. Increased study time through practice outside the classroom
2. Spaced learning through reviews outside the classroom
3. Students' L2 proficiency

Study 1

4. Cumulative testing works in aural modality?

Study 2

- 1) Does cumulative testing increase study time and/or review counts?
 - Cumulative > Noncumulative for both
- 2) Do increased study time and/or review counts moderate the benefit of cumulative testing?
 - Cumulative > Noncumulative but only with practice
- 3) How study time, review counts, and L2 proficiency together moderate the benefit of cumulative testing?
 - Cumulative > Noncumulative but only with practice

Overview

Study 1: Cumulative vs. Noncumulative - Written

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Participants

- cumulative: $n = 41$
- noncumulative: $n = 43$

Target words ($k = 100$)

- 100 words from TOEFL books

Weekly Quiz ($k = 10$ each)

- 5 meaning recall
- 5 form recall

Pre/Posttest ($k = 100$ each)

- 50 meaning recall (A, B List)
- 50 form recall (A, B List)

Week	Word List	Cumulative	Noncumulative	
1		Pretest	Pretest	2 weeks
2	List 1			
3	List 2	Quiz 1 (List 1)	Quiz 1 (List 1)	10 weeks
4	List 3	Quiz 2 (List 1–2)	Quiz 2 (List 2)	
5	List 4	Quiz 3 (List 1–3)	Quiz 3 (List 3)	
6	List 5	Quiz 4 (List 1–4)	Quiz 4 (List 4)	
7	List 6	Quiz 5 (List 1–5)	Quiz 5 (List 5)	
8	List 7	Quiz 6 (List 1–6)	Quiz 6 (List 6)	
9	List 8	Quiz 7 (List 1–7)	Quiz 7 (List 7)	
10	List 9	Quiz 8 (List 1–8)	Quiz 8 (List 8)	
11	List 10	Quiz 9 (List 1–9)	Quiz 9 (List 9)	
12		Quiz 10 (List 1–10)	Quiz 10 (List 10)	
13				3 weeks
14				
15		Posttest	Posttest	

Word List 1

N = Noun V = Verb Adj = Adjective Adv = Adverb

adolescent [Adj] (思春期の)

- in the process of developing from a child into an adult
- *The study focused on the eating habits of adolescent girls.*

approximately [Adv] (およそ; 約)

- almost, but not completely, accurate or exact; roughly.
- *The small town has approximately 2,000 inhabitants.*

boundary [N] (境界; 境界線)

- a line that marks the limits of an area; a dividing line.
- *The river forms the natural boundary between the two countries.*

decimal [N, Adj] (小数点; 小数点の)

- relating to or denoting a system of numbers based on the number ten, tenths, and powers of ten.
- *Please list the values after the decimal point.*

diplomacy [N] (外交)

- the profession, activity, or skill of managing international relations, typically by a country's representatives abroad.
- *Diplomacy is essential in preventing conflicts between nations.*

ログイン画面

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(A)

English I-A: Word Cards

[Word List 1 \(test: 4/23\)](#)

[Word List 2 \(test: 4/30\)](#)

[Word List 3 \(test: 5/7\)](#)

[Word List 4 \(test: 5/14\)](#)

[Word List 5 \(test: 5/21\)](#)

[Word List 6 \(test: 5/28\)](#)

[Word List 7 \(test: 6/4\)](#)

[Word List 8 \(test: 6/11\)](#)

[Word List 9 \(test: 6/18\)](#)

[Word List 10 \(test: 6/25\)](#)

(B)

adolescent

(C)

思春期の

(D)

Study Time:

- overall practice counts

Review Counts:

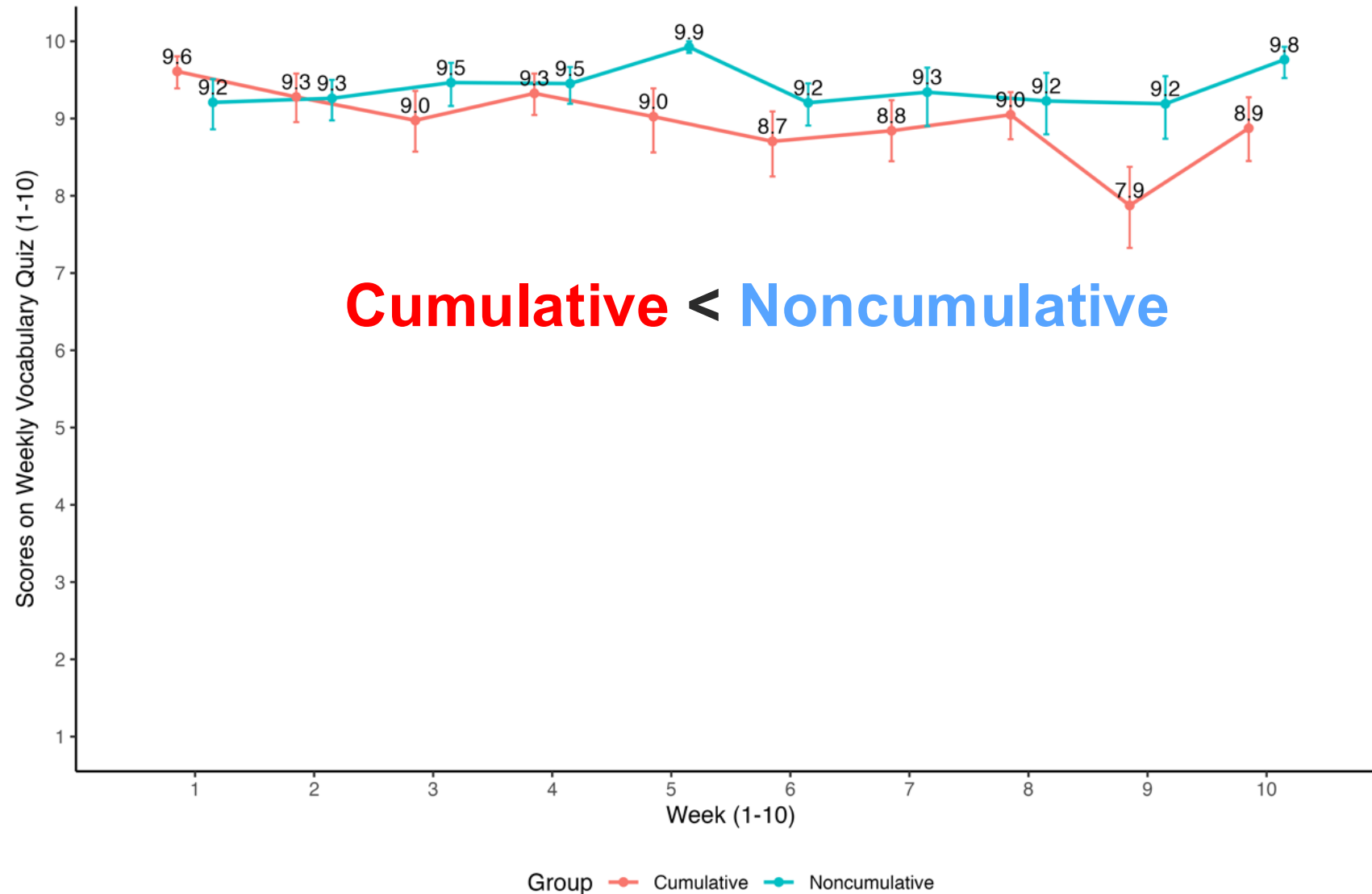
- review counts after the quiz week

Results

Results: Weekly Quiz

Study 1: Cumulative vs. Noncumulative - Written

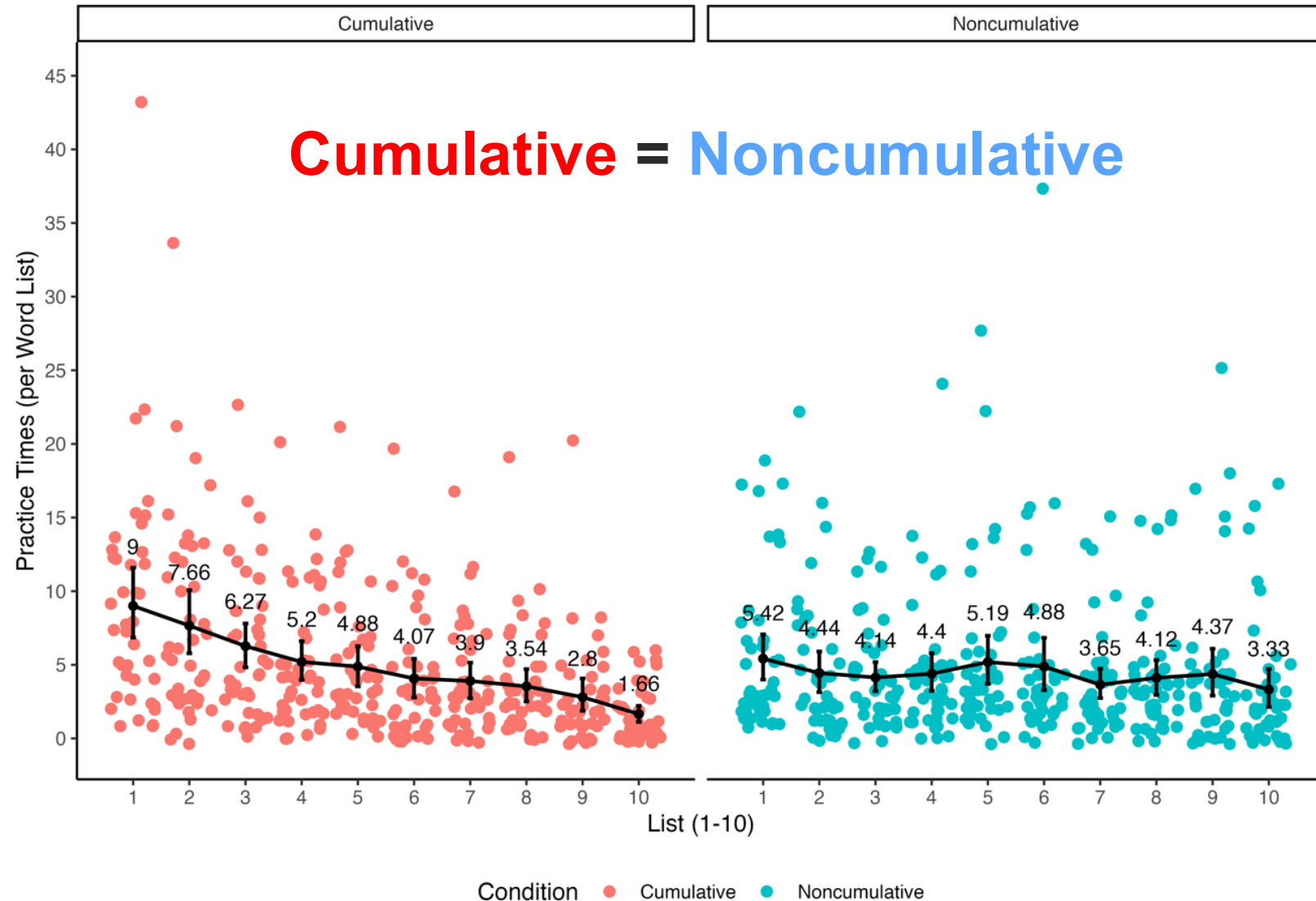
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Results: Practice Counts

Study 1: Cumulative vs. Noncumulative - Written

14

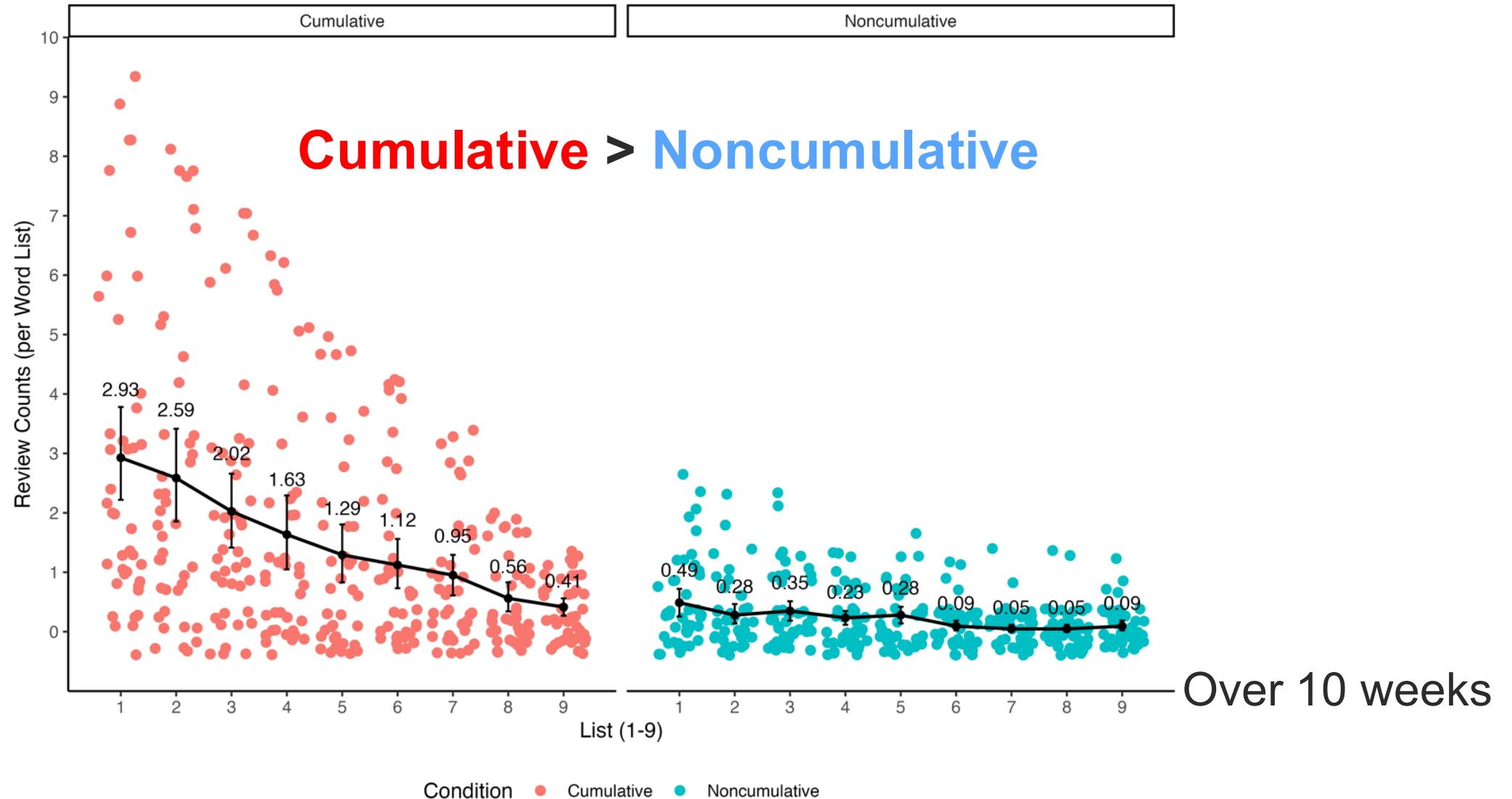


Over 10 weeks

Results: Review Counts

Study 1: Cumulative vs. Noncumulative - Written

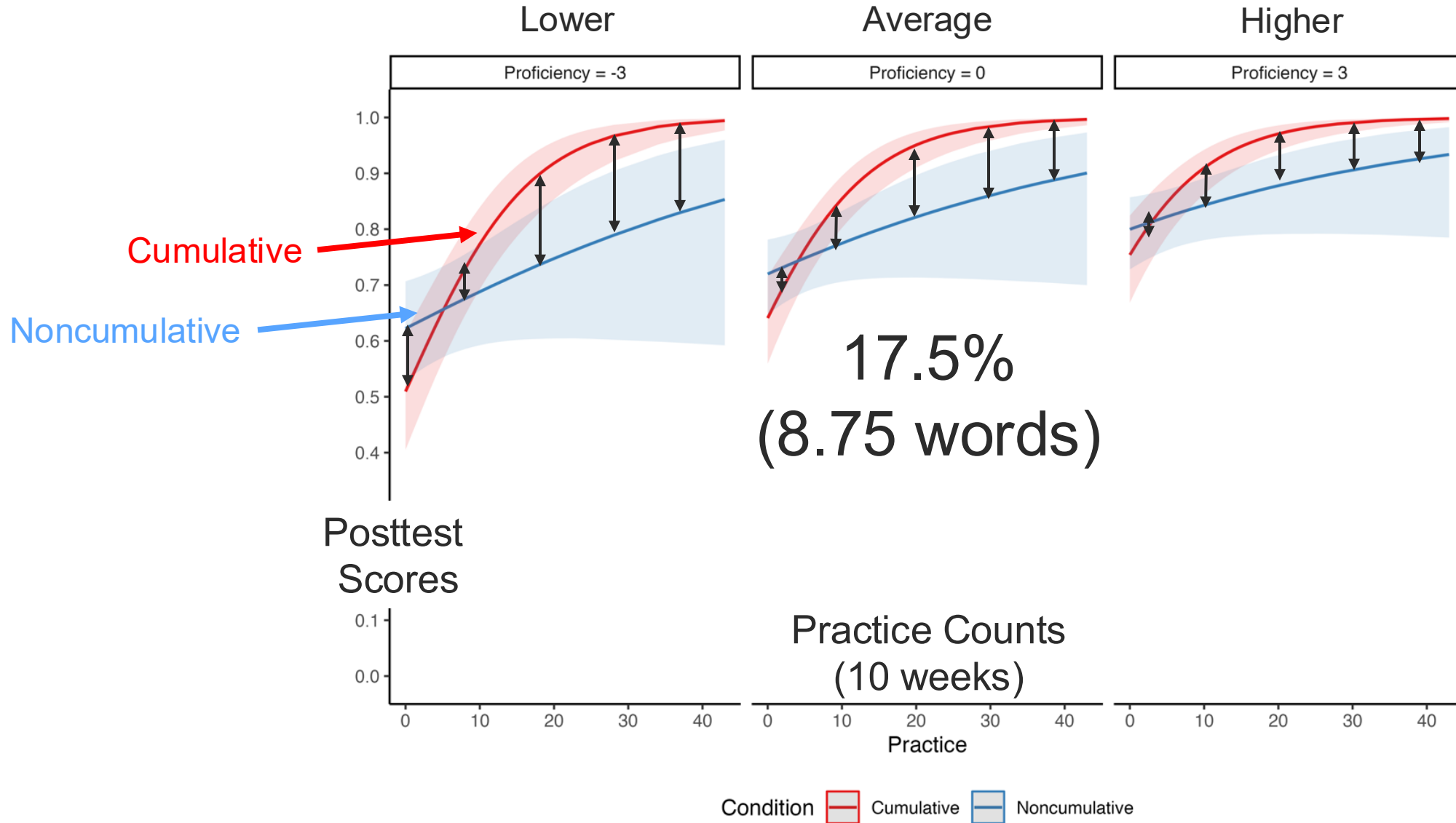
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Results: Posttests (Meaning Recall)

Study 1: Cumulative vs. Noncumulative - Written

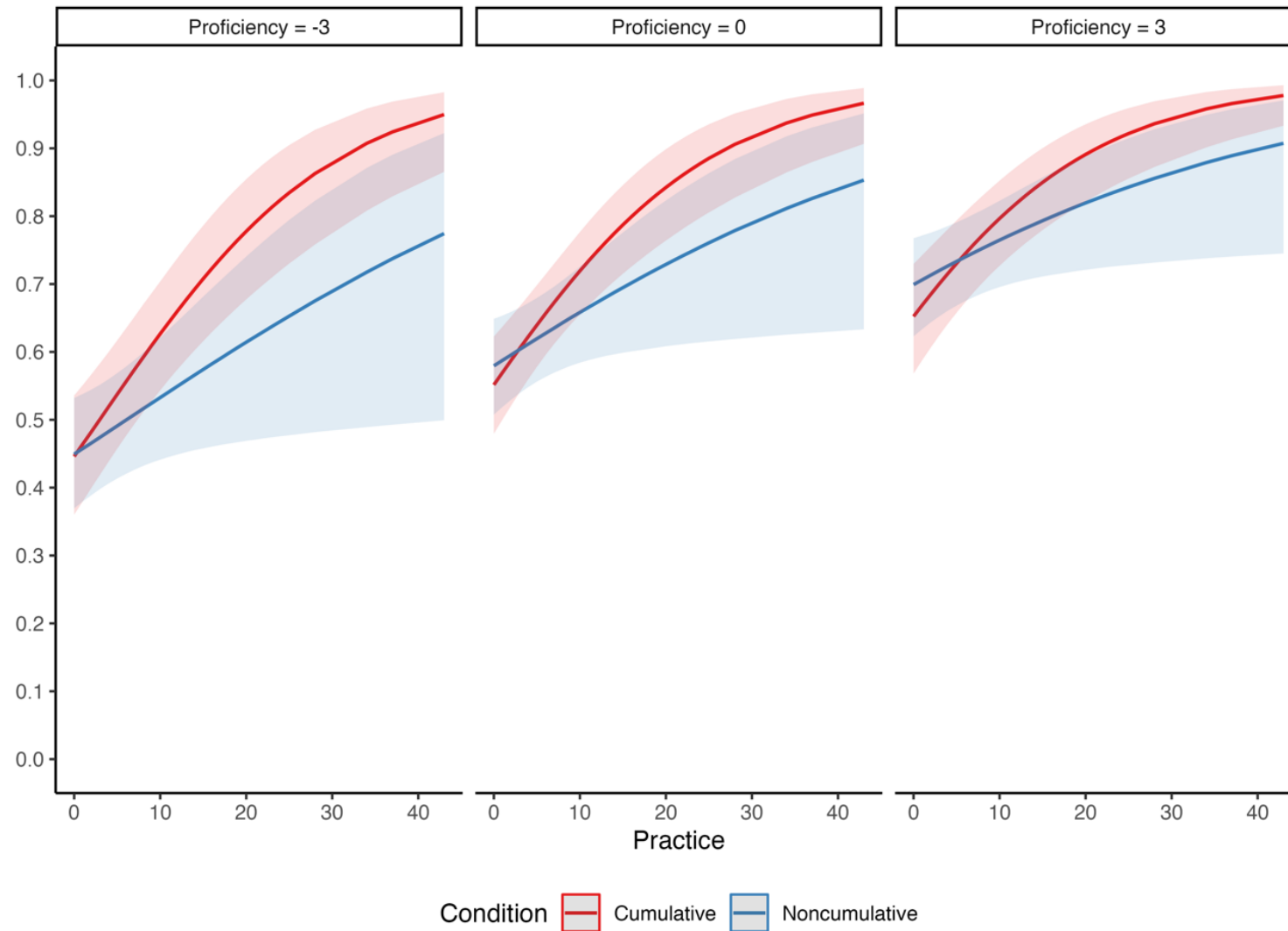
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Results: Posttests (Form Recall)

Study 1: Cumulative vs. Noncumulative - Written

17

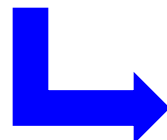


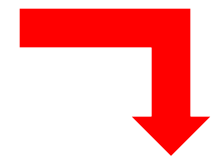
Summary of Study 1

Study 1: Cumulative vs. Noncumulative - Written

18

- Cumulative is superior to noncumulative testing
- The benefit only when there is practice
- Increasing practice means increasing benefit
- Review is more important than study time
- The benefit is larger for those with lower proficiency

 **Written** vocabulary knowledge
acquired from **written** practice

 **Spoken** vocabulary knowledge
acquired from **spoken** practice



Participants

- cumulative: $n = 39$
- noncumulative: $n = 43$

Target words ($k = 100$)

- 100 words from TOEFL books

Weekly Quiz ($k = 10$ each)

- 10 **spoken** meaning recall
(Writing meaning upon hearing sounds)

Pre/Posttest ($k = 100$ each)

- **Spoken** meaning recall

Week	Word List	Cumulative	Noncumulative	
1		Pretest	Pretest	2 weeks
2	List 1			
3	List 2	Quiz 1 (List 1)	Quiz 1 (List 1)	10 weeks
4	List 3	Quiz 2 (List 1–2)	Quiz 2 (List 2)	
5	List 4	Quiz 3 (List 1–3)	Quiz 3 (List 3)	
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11	List 10	Quiz 9 (List 1–9)	Quiz 9 (List 9)	
12		Quiz 10 (List 1–10)	Quiz 10 (List 10)	
13				3 weeks
14				
15		Posttest	Posttest	

ログイン画面 **Log in page**

Student ID 学籍番号:

Password パスワード:

Log in

(A)

English II-A: Word Cards

- Word List 1

[Written](#) [Auditory](#)

- Word List 2

[Written](#) [Auditory](#)

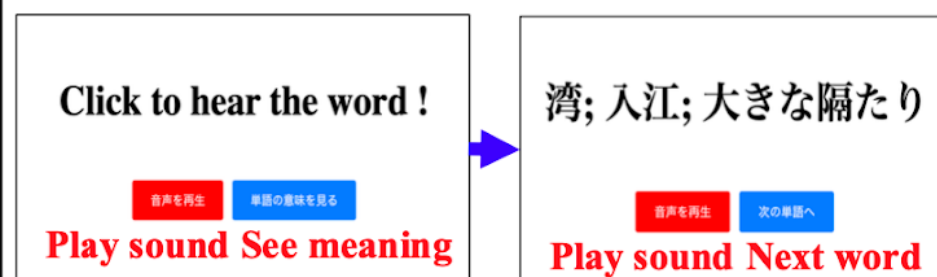
- Word List 3

[Written](#) [Auditory](#)

(B)



(C)



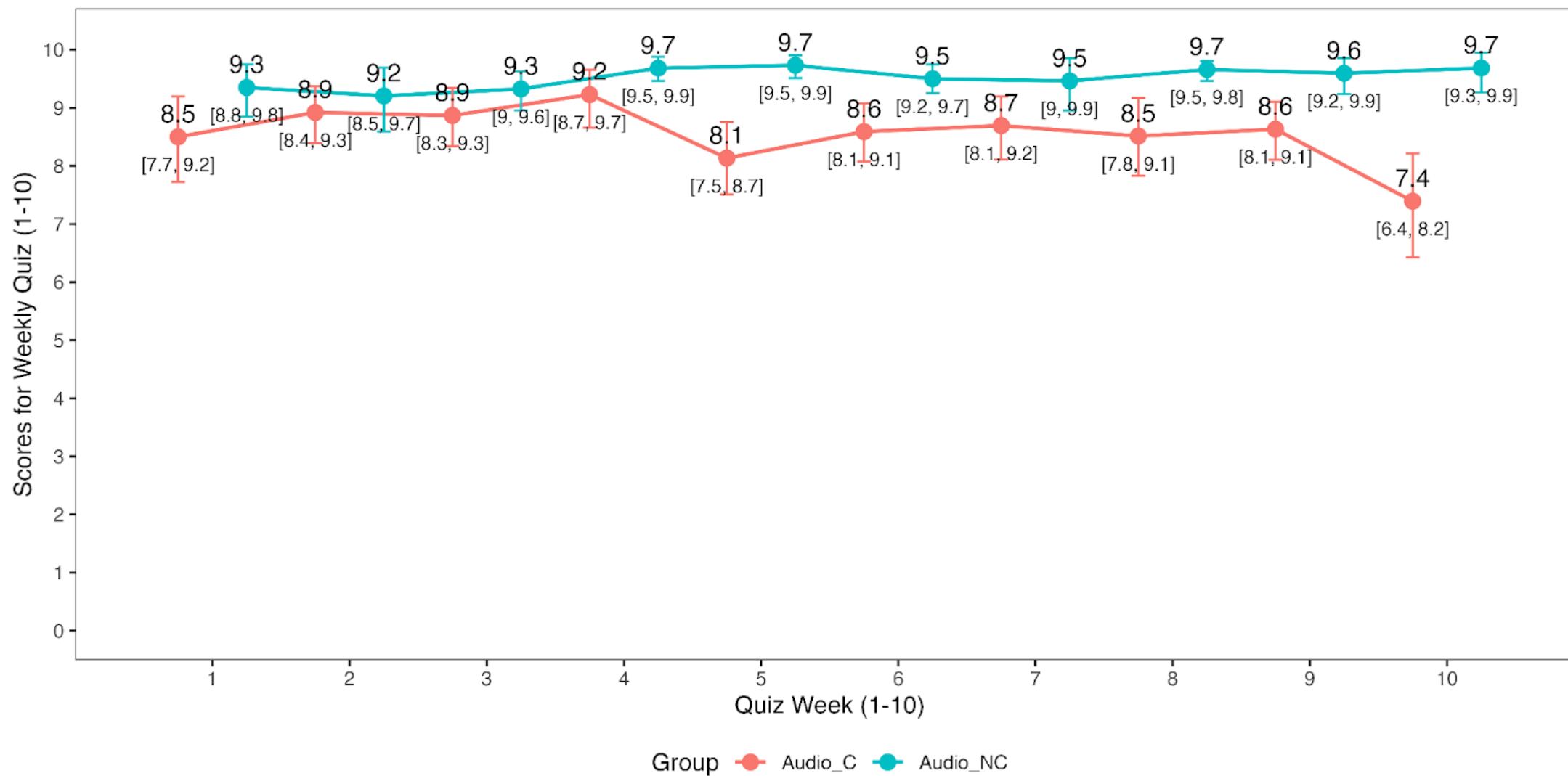
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Results

Results: Weekly Quiz

Study 2: Cumulative vs. Noncumulative - Spoken

22



Results: Practice Behaviors

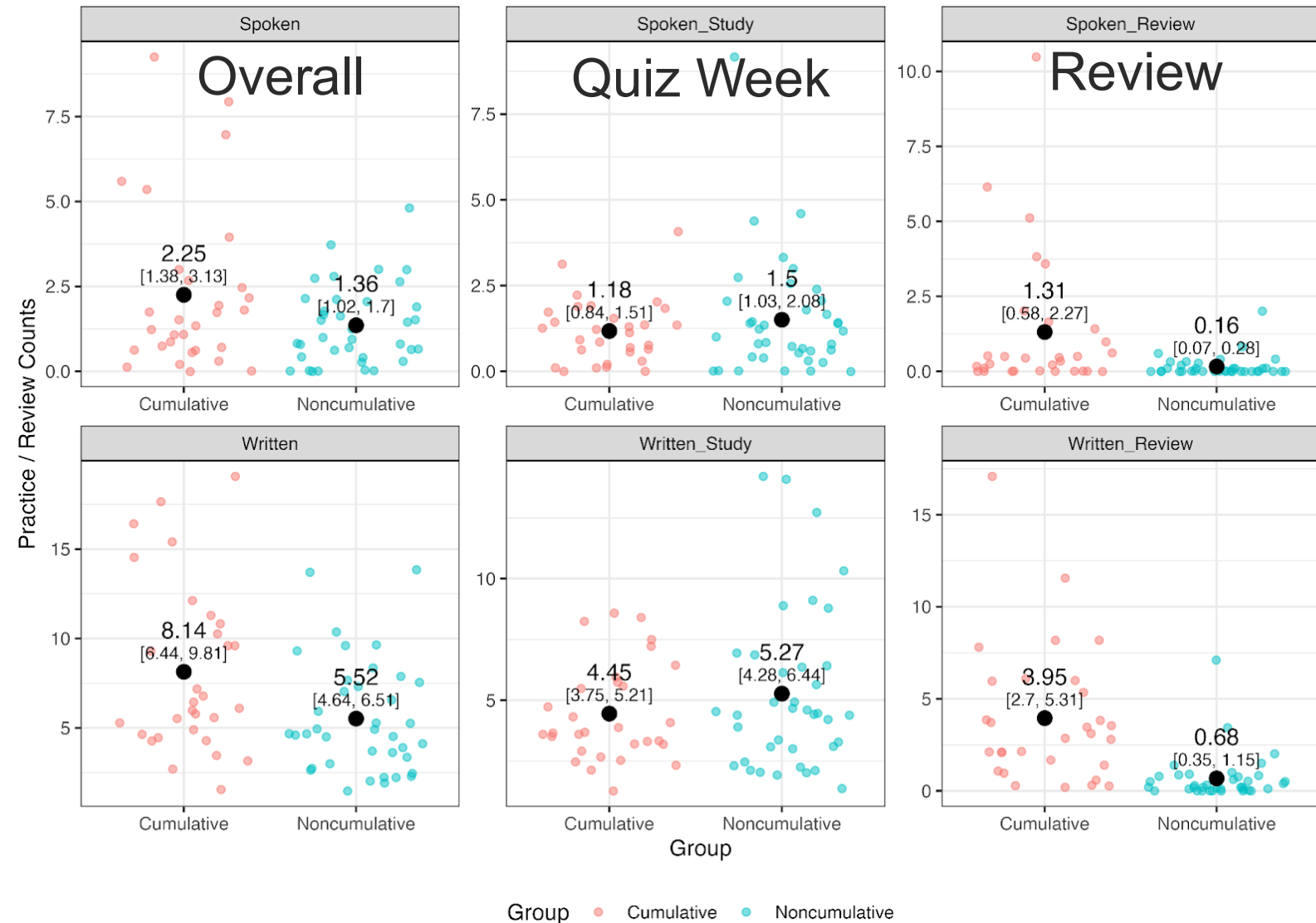
Study 2: Cumulative vs. Noncumulative - Spoken

23

Spoken practice

VS.

Written practice

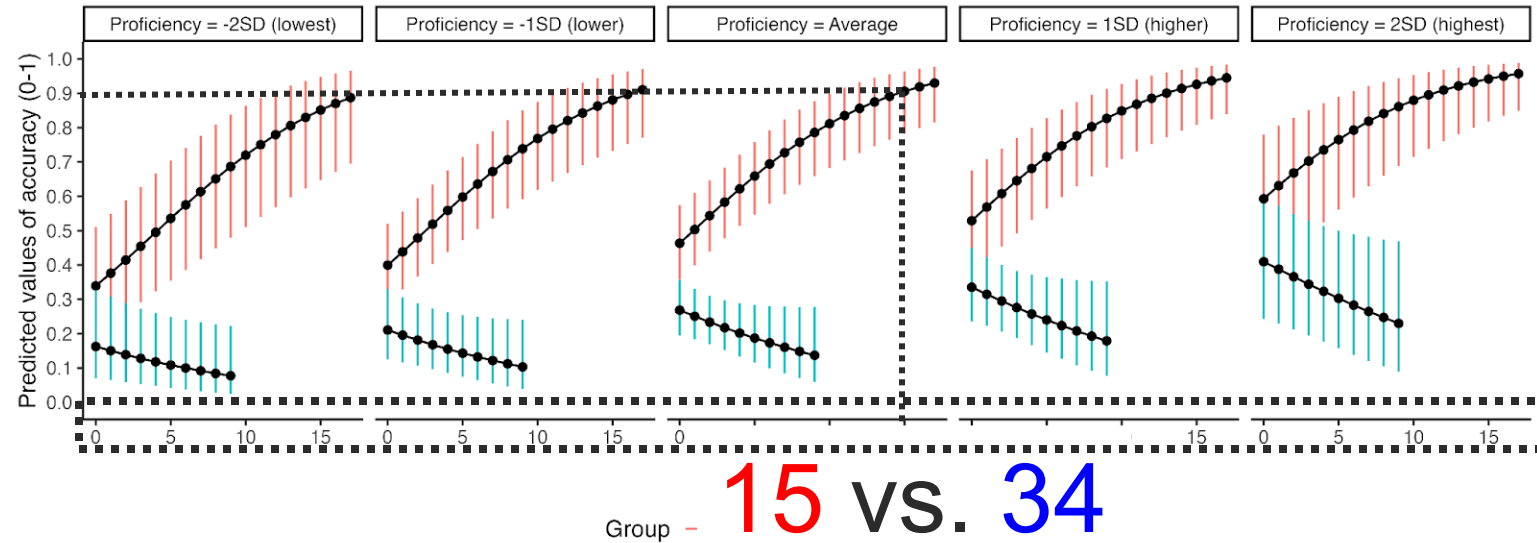


Results: Posttest

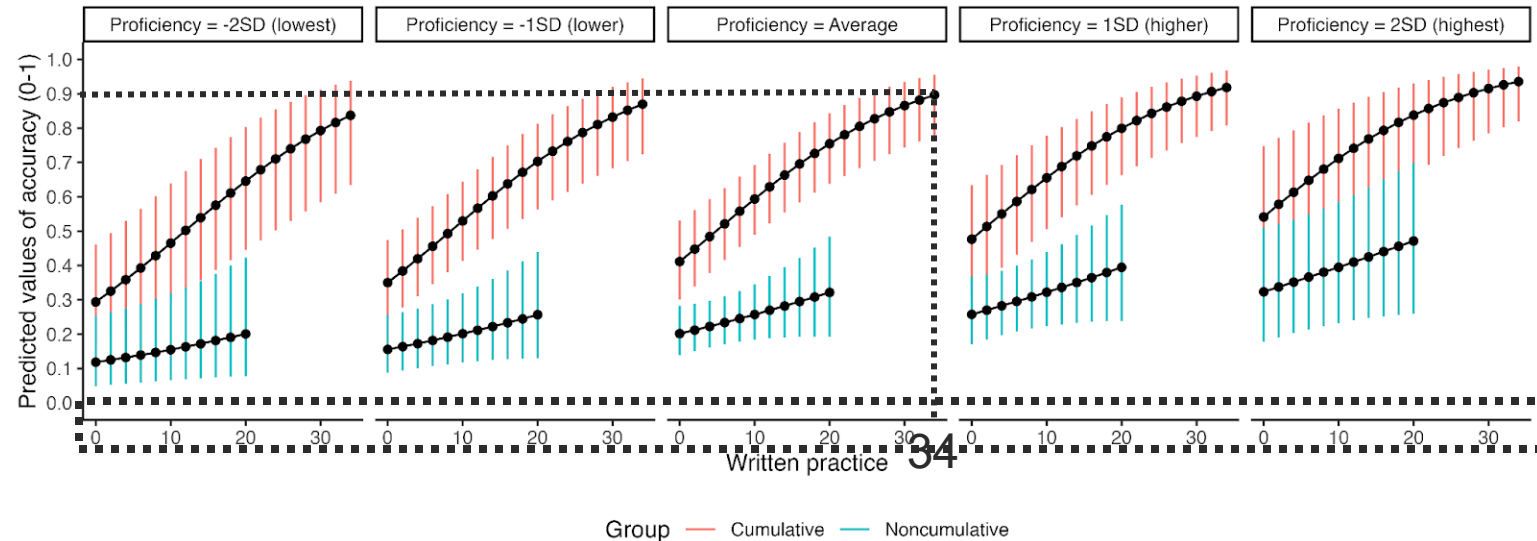
Study 2: Cumulative vs. Noncumulative - Spoken

24

Spoken practice



Written practice



Summary of Study 2

Study 2: Cumulative vs. Noncumulative - Spoken

25

- Cumulative is superior to noncumulative testing
- Students prefer **written** practice even when the quiz is a **spoken** test
- But(!) **spoken** vocab knowledge requires **spoken** practice
- What happens when we compare:
 - **spoken** cumulative vs. **written** cumulative
 - opportunities for **spoken** and **written** practice
 - acquisition of both **spoken** and **written** knowledge

What if we compare spoken vs. written cumulative testing?

- Providing both **spoken** and **written** practice opportunities?
 - Which practice do students prefer - **spoken** or **written** practice or **both**?
- Testing both **spoken** and **written** meaning recall?
 - Which group is better than which knowledge
 - Which practice type predicts which knowledge for which group?

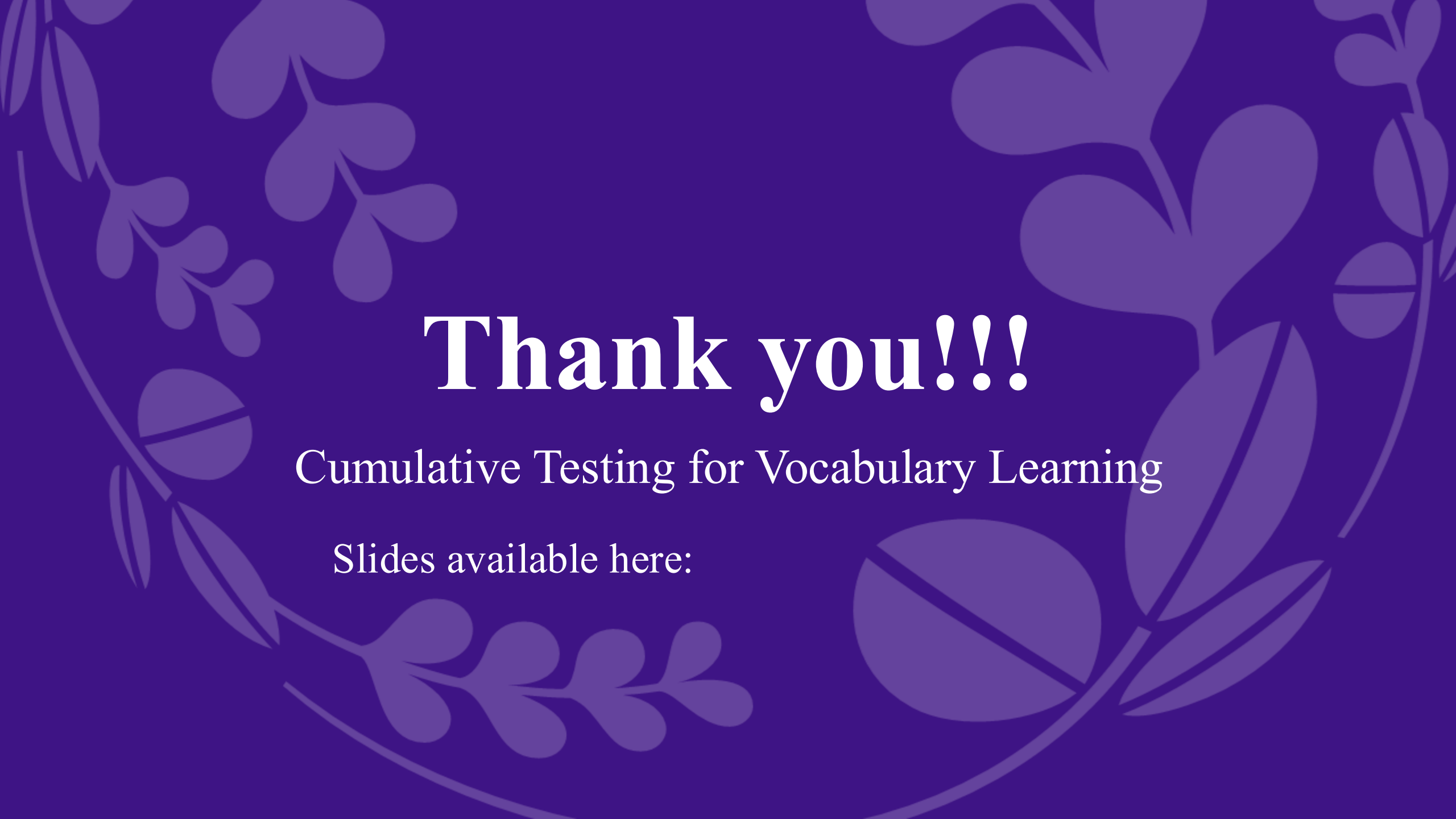
Implications for teaching vocabulary

Discussion and pedagogical implications

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- Cumulative tests work for both spoken and written vocabulary learning
- The benefit holds only when students practice
 - Just testing is not helpful
- Spaced learning (reviews) is more critical than mere study time
- Practice in the respective mode is necessary
 - transfer-appropriate practice!
- Teacher's job: encourage students to encounter spoken words

- Comparing written vs. spoken CTs (Uchihara, Maie, Komuro, & Oikawa, in progress)
- Investigating the optimal spacing for review (data available)
- Add a talker-variability component (Wang for MA project)
- Different forms of practice (e.g., L1→L2 vs. L2→L1)
- Student modeling
 - increasing complexity after criterion
 - active vs. passive / recall vs. recognition



Thank you!!!

Cumulative Testing for Vocabulary Learning

Slides available here: